31. Construct a C Program to Simulate the First in First Out (FIFO) Paging Technique of Memory Management

Code - Part 1 (Lines 1–35)

```c
1  #include <stdio.h>
2
3  #define MAX_FRAMES 3
4
5  // Function to print frames
6  void printFrames(int frames[], int n) {
7      for (int i = 0; i < n; i++) {
8          if (frames[i] == -1)
9              printf(" - ");
10         else
11             printf(" %d ", frames[i]);
12     }
13     printf("\n");
14 }
15
16 int main() {
17
18     int referenceString[] = {7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2};
19     int n = sizeof(referenceString) / sizeof(referenceString[0]);
20
21     int frames[MAX_FRAMES];
22     int framePointer = 0;
23
24     // Initialize frames
25     for (int i = 0; i < MAX_FRAMES; i++) {
26         frames[i] = -1;
27     }
28
29     printf("Reference String: ");
30     for (int i = 0; i < n; i++) {
31         printf("%d ", referenceString[i]);
32     }
33
34     printf("\n\nPage Replacement Order:\n");
35
```

Code - Part 2 (Lines 36–61)

```
36      for (int i = 0; i < n; i++) {
37
38          int page = referenceString[i];
39          int pageFound = 0;
40
41          // Check if page exists
42          for (int j = 0; j < MAX_FRAMES; j++) {
43              if (frames[j] == page) {
44                  pageFound = 1;
45                  break;
46              }
47          }
48
49          // If not found → replace
50          if (!pageFound) {
51              printf("Page %d -> ", page);
52
53              frames[framePointer] = page;
54              framePointer = (framePointer + 1) % MAX_FRAMES;
55
56              printFrames(frames, MAX_FRAMES);
57          }
58      }
59
60      return 0;
61  }
```

Output

```
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2

Page Replacement Order:
Page 7 ->  7  -  -
Page 0 ->  7  0  -
Page 1 ->  7  0  1
Page 2 ->  2  0  1
Page 3 ->  2  3  1
Page 0 ->  2  3  0
Page 4 ->  4  3  0
Page 2 ->  4  2  0
Page 3 ->  4  2  3
Page 0 ->  0  2  3


...Program finished with exit code 0
Press ENTER to exit console.
```